# Assessment Tracks Specification

Separate demand tracks · Garbage unit structures · Monthly verification before bill generation

## 1. Separate assessment tracks for every head

Each revenue head is its own independent assessment / demand track. A change under one head never alters the demand or unit structure of another head on the same parent property.

| Head | Frequency | Unit-level? | Assessment artefact |
|---|---|---|---|
| Property Tax | Yearly (FY) | No – building level | Property Tax Assessment Sheet |
| **Garbage (Solid Waste)** | Monthly or Yearly | **Yes – per Res / Com unit** | **Garbage Assessment Sheet** |
| Water | Monthly or Yearly | No – connection level | Water Assessment Sheet |
| Sewerage | Monthly or Yearly | No – connection level | Sewerage Assessment Sheet |
| Other | Monthly or Yearly | Configurable | Other Assessment Sheet |

### Core principles

• **One parent property, many independent tracks.** GIS survey is the single source of identity. Each head keeps its own assessment, rates, demand ledger and collection history.

• **Garbage is unit-billed.** Residential and Commercial units live under the parent. Each active unit can generate its own garbage demand. Buildings without a unit register keep legacy single-key behaviour.

• **Independent updates.** Adding/removing a garbage unit recalculates only Garbage demand. Property Tax, Water and Sewerage on the same property stay untouched.

• **Existing code.** `units` module, residential/commercial rate slots, ledger keys `garbage~{propertyId}~{unitId}`.

## 2. Monthly verification BEFORE generating the bill (required)

Every month, before the Demand Centre raises Garbage demand (and similarly for other monthly heads), the system must show a verification screen:

1. **List of structures** in the town, each with its units shown **name-wise** (label + occupant).
2. Units grouped as **Residential** and **Commercial**, with occupancy (Owner / Tenant / Vacant) and active flag.
3. **Option to Add** a new Residential or Commercial unit on that structure (label, type, occupant).
4. **Option to Remove** an existing unit (soft-remove: set active = false; history kept).
5. Only after the admin clicks **“Units verified — Generate demand”** is the monthly demand written. A demand cannot be edited afterwards; only adjusted or reversed.

*This verification is enforced in the Demand Centre for unit-billed heads (Garbage). Other monthly heads still use the confirm step; the same name-wise list pattern can be extended to them.*

# 3. Garbage Assessment Sheet + verification UI (sample)

Property identity block and the unit register that the verification modal edits.

| | | | |
|---|---|---|---|
| **ULB** | Municipal Council Shimla | **Ward** | Ward 07 – Lower Bazaar |
| **Property ID** | SMC-07-0142 | **Owner** | Sh. Ramesh Kumar |
| **Address** | Shop No. 12–14, Ground & First Floor, Cart Road, Shimla – 171001 | **Use** | Mixed |

## Units / Structures — name-wise list (verification + assessment)

| # | Label (structure name) | Type | Occupancy | Occupant | Active | Action |
|---|---|---|---|---|---|---|
| U01 | Ground Floor — Shop 1 | Commercial | Tenant | M/s Himalaya Traders | Yes | Remove |
| U02 | Ground Floor — Shop 2 | Commercial | Owner | Sh. Ramesh Kumar | Yes | Remove |
| U03 | First Floor — Flat A | Residential | Owner | Sh. Ramesh Kumar | Yes | Remove |
| U04 | First Floor — Flat B | Residential | Vacant | — | Yes | Remove |

[ + Add Residential unit ] [ + Add Commercial unit ] [ Save units ]

| Component | Count | Rate (Rs) | Amount (Rs) |
|---|---|---|---|
| Residential units (active) | 2 | 50 | 100 |
| Commercial units (active) | 2 | 120 | 240 |
| **Total Garbage Demand (this period)** | | | **340** |

*Demand Centre flow: Generate Garbage → verification modal opens → admin reviews name-wise list, adds/removes units, saves → "Units verified — Generate demand" writes the period demand.*

## 4. Who performs each function + structure rules

| Function | Primary role | Notes |
|---|---|---|
| GIS Property Survey | Field Surveyor | Geometry, owner, address; may seed initial units. |
| Property Tax Assessment | ULB Admin / Assessor | Draft → Supervisor approval. Independent of garbage units. |
| **Garbage Unit Structure (add/remove Res & Com)** | **Field Surveyor / ULB Admin** | Units page or Demand Centre verification modal. Soft-remove keeps history. |
| **Monthly unit verification before demand** | **ULB Admin** | Mandatory for Garbage each month. Name-wise list + add/remove then generate. |
| **Garbage Assessment & Demand** | **ULB Admin / Assessor** | Only after verification. Applies Res/Com rates to active unit counts. |
| Garbage Collection (Counter) | Counter Operator | Payment against garbage ledger (per unit or rolled up). |
| Water / Sewerage Collection | Counter Operator | Separate ledgers; not linked to garbage unit register. |
| User & Role Management | ULB Admin | Creates surveyor, assessor, cashier, public accounts. |

### Rules for add / remove

1. **Add** – Choose Residential or Commercial, enter label and occupant. System assigns next unit id (U01, U02 …). Active by default.
2. **Remove** – Sets active = false. Does not delete. Historical demand/receipts stay visible. Future demand ignores inactive units.
3. **Type change** – Residential ↔ Commercial moves the unit to the other rate bucket for the next demand.
4. **Mixed buildings without units** – Cannot be billed for garbage until at least one unit is added.
5. **Independence** – All unit operations affect only the Garbage track.

### Implementation note (code)

Demand Centre: Generate on a unit-billed head opens `UnitVerifyModal`. The modal lists every structure with name-wise units, supports add/remove + Save, then “Units verified — Generate demand” calls the existing demand generator. Units page (`UnitsManager`) remains available for bulk editing outside the monthly cycle.

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*Matches existing app concepts (units module, Res/Com rate slots) and formalises monthly name-wise verification with add/remove before any Garbage bill is generated.*